

◆ M001 Week 7 — English Worksheet (Beginner)

Topic: Pyramid Variations · **Course:** Pyramid Problems · **Level:** Beginner · **Time:** about 30 minutes

This week you change **one thing** about your pyramid to make a new shape. A **hollow** layer only has blocks on the edge. A **stepped** pyramid shrinks faster, so the steps look bigger.

1 · Predict 🧠

Read the steps. Before you imagine the agent doing it, circle or write your answer.

```
set size to 5
while size > 0:
    [build one HOLLOW size × size layer]
    move up by 1
    set size to size - 2
```

Is this pyramid filled inside or empty inside? Circle one: filled · empty

Could a person stand inside it?


```
set size to 9
while size > 0:
    [build one size × size square layer]
    move up by 1
    set size to size - 4
```

A normal pyramid shrinks by 2. This one shrinks by 4. Do the steps look bigger or smaller?

Circle one: bigger · smaller

2 · Find the Difference 🐛

Each pair shows clean steps first, then a broken version of the same idea. Circle what's different and write one short sentence about the bug.

Pair A — The layer should be **hollow** — blocks only on the edge.

```
# clean
[build one HOLLOW layer]
```

```
# buggy
[build one FULL layer]
```

What is wrong? What does the buggy pyramid look like inside?

Pair B — The stepped pyramid should shrink by **4** each layer.

```
# clean
while size > 0:
    [build layer]
    move up by 1
    set size to size - 4
```

```
# buggy
while size > 0:
    [build layer]
    move up by 1
    set size to size - 2
```

What is wrong? Which pyramid has more layers?

3 · Fill the Gap

You want a **stepped** pyramid that shrinks fast. One number is missing. Fill it in using the word bank.

```
set size to 9
while size > 0:
    [build layer]
    move up by 1
    set size to size - ____
```

Word bank: 4 · 0 · 100

Write the missing number:

4 · Tell Me What You Built 📷

Now switch to your homework world. Pick **one** variation and build it:

- **Hollow** — layers with blocks only on the edge, or
- **Stepped** — shrink by 4 instead of 2.

When you finish, come back here. Send a photo or video of your variation, then explain what you did. Use these sentence starters — write 2 or 3 sentences.

I picked the ... variation.

I changed the ... part of the code.

The hardest part was ...

5 · Record Your Walkthrough

Take a video on your phone (or a parent's phone) and walk the camera around your variation. Talk like you are teaching a friend. Try to use these words: **hollow**, **stepped**, **layer**, **edge**.

1. Show what makes your pyramid different from a normal one.
2. Say which part of the code you changed.
3. Show one bug you hit and how you fixed it.

Write what you will say in your video. You can read from it while filming.

Submit 

Send this worksheet + your walkthrough video to teacher on KakaoTalk.